Chapter 2

The setting: a discrete-time market with frictions and convex risk measures

This chapter sets up all objects needed in the rest of the thesis. We first construct the market: trading dates, information, hedging instruments, trading strategies, constraints and transaction costs. Then we use these objects to define the terminal profit-and-loss equation. The second half of the chapter develops convex risk measures, and in particular the class of optimized certainty equivalents, to measure the "goodness" of the P&L.

The market model (Sections 2.1–2.2) copies the setup of Bühler et al. [2, Section 2], with one structural difference: the probability space is *general*, not finite.

2.1 The market

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We emphasize that $(\Omega, \mathcal{F}, \mathbb{P})$ is a general probability space: in contrast to Bühler et al. [2], we do not assume that Ω is finite. All identities and inequalities between random variables are understood in the $\mathbb{P}$-almost sure sense.

Trading takes place at finitely many dates

$$0 = t_0 < t_1 < \cdots < t_n = T.$$

New market information available at t_k is modelled by an $\mathbb{R}^r$-valued random vector I_k; the vector I_k may contain market prices of the hedging instruments, but also additional signals such as prices of non-traded instruments, news or trading volumes. The flow of information is described by the filtration $\mathbb{F} = (\mathcal{F}_k)_{k=0,\dots,n}$ generated by the *information process* $I = (I_k)_{k=0,\dots,n}$,

$$\mathcal{F}_k := \sigma(I_0, \dots, I_k), \qquad k = 0, \dots, n,$$

and we set $\mathcal{F}_T := \mathcal{F}_n$. Without loss of generality we take $\mathcal{F} = \mathcal{F}_T$.

The market contains d hedging instruments whose *mid-prices* are given by an $\mathbb{R}^d$-valued $\mathbb{F}$-adapted process $S = (S_k)_{k=0,\dots,n}$, where S_k is the price vector at date t_k. The instruments may be primary assets such as stocks or currencies, but also secondary assets such as liquidly traded options; this flexibility is one of the strengths of the framework, because hedging a complex derivative portfolio in practice relies on liquid vanilla options as much as on the underlying itself. We stress that S is simply an adapted process: no martingale-measure assumption of any kind is made at this point.

The *liability* is an $\mathcal{F}_T$-measurable random variable Z, interpreted as the payoff of a portfolio of derivatives that the trader has sold and must service at time T. As in Bühler et al. [2, Section 2], we adopt the following simplifications. Interest rates are zero, and any cash flow before T is assumed to be invested into the overnight risk-free account. Under zero rates, this is costless, so we can assume all payments are due at the terminal date T. We also exclude American-style optionality.

2.2 Trading strategies, costs and the P&L equation

A *trading strategy* is an $\mathbb{R}^d$-valued $\mathbb{F}$-adapted process $\delta = (\delta_k)_{k=0,\dots,n-1}$, where δ_k^i is the position in the i-th hedging instrument held over the interval $(t_k, t_{k+1}]$. Adaptedness means that the position chosen at t_k may only use information available at t_k. We use throughout the convention

$$\delta_{-1} := 0, \qquad \delta_n := 0:$$

the trader starts with no position and liquidates the entire portfolio at T. We write $\mathcal{H}^{\mathrm{u}}$ for the set of all such strategies (the superscript stands for "unconstrained").

In real markets, not every position is available: instruments may be tradeable only from certain dates onwards, liquidity limits the position size and risk management imposes bounds or short-selling restrictions. Following Bühler et al. [2, Section 2.4], such *trading constraints* are encoded by continuous maps H_k that project an intended position onto the set of admissible ones, in a way that depends on the information available up to t_k and on the previously held position δ_{k-1}. The constrained version of $\delta \in \mathcal{H}^{\mathrm{u}}$ is denoted by $H \circ \delta$ and defined recursively by

$$(H \circ \delta)_k := H_k\big((H \circ \delta)_0, \dots, (H \circ \delta)_{k-1}, \delta_k\big), \qquad k = 0, \dots, n-1,$$

with $H_k(\cdot, 0) = 0$, so that the zero strategy is always admissible. The set of admissible strategies is

$$\mathcal{H} := \{H \circ \delta : \delta \in \mathcal{H}^{\mathrm{u}}\}.$$

In Chapters 3 and 4 we will restrict admissibility by requiring that strategies cannot trade into a loss.

Trading is costly. For each k, the *cost function* $c_k \colon \Omega \times \mathbb{R}^d \to [0, \infty)$ gives the cost $c_k(x)$ of changing the position by $x \in \mathbb{R}^d$ at date t_k; it is $\mathcal{F}_k \otimes \mathcal{B}(\mathbb{R}^d)$-measurable, upper semicontinuous in the spatial variable and normalized so that

$$c_k(0) = 0 :$$

not trading is free and costs are nonnegative. The *total cost* of a strategy δ is

$$C_T(\delta) := \sum_{k=0}^{n} c_k(\delta_k - \delta_{k-1}),$$

where the conventions $\delta_{-1} = \delta_n = 0$ ensure that the initial build-up and the final liquidation of the position are charged as well.

Example 2.1. Following Bühler et al. [2, Section 2], we give an example of proportional transaction costs that given by

$$c_k(x) = \sum_{i=1}^{d} \kappa^i S_k^i |x^i|$$

with relative cost parameters $\kappa^i \geq 0$: each trade costs a fixed proportion of the transacted volume. These cost functions are continuous, convex and normalized.

Upper semicontinuity of the cost functions is exactly what the limit arguments of Chapter 4 require: along a convergent sequence of strategies, costs may drop in the limit but not jump up.

The trading gains of δ from price movements are given by the discrete stochastic integral

$$(\delta \cdot S)_T := \sum_{k=0}^{n-1} \delta_k \cdot (S_{k+1} - S_k).$$

Now suppose the trader sells the liability Z for an initial cash amount p_0 and trades according to $\delta \in \mathcal{H}$. Collecting all cash flows at T yields the *terminal profit-and-loss* of Bühler et al. [2, Equation (1)],

$$\mathrm{PL}_T(Z, p_0, \delta) := -Z + p_0 + (\delta \cdot S)_T - C_T(\delta). \tag{2.1}$$

Equation (2.1) is the central object of this thesis. Its four terms are the payment of the liability, the premium received, the trading gains and the frictional losses. The remainder of the thesis is organized around the question of how to determine the riskiness of the random variable (2.1), how to choose δ (Chapter 3) and to compute such a δ (Chapter 4) so as to minimize its risk as much as possible.

2.3 Relevant markets and feasibility

It is convenient to isolate the part of the P&L that is generated by pure trading. For $\delta \in \mathcal{H}$, define the *gain* of δ net of costs by

$$W(\delta) := (\delta \cdot S)_T - C_T(\delta).$$

A first natural question is whether the market allows arbitrage: a strategy $\delta \in \mathcal{H}$ with

$$W(\delta) \geq 0 \ \mathbb{P}\text{-a.s.} \qquad \text{and} \qquad \mathbb{P}[W(\delta) > 0] > 0.$$

Classical pricing theory begins by excluding such strategies. The deep hedging framework does not: no no-arbitrage assumption is imposed a priori. This is a deliberate modelling choice. The framework is designed to operate on market data or on scenarios from a simulator, and neither is guaranteed to be arbitrage-free. What must be avoided is not arbitrage itself but a market that is *too* attractive, in the following sense.

Once a risk functional ρ is fixed (Sections 2.4–2.6 below construct it), the quantity

$$\pi(0) := \inf_{\delta \in \mathcal{H}} \rho\big(W(\delta)\big)$$

measures the risk of the position of a trader with no liability who is free to trade. A proper definition and the systematic study of this optimization problem is the subject of Chapter 3. Following Bühler et al. [2, Section 3.1], the market is called *irrelevant* for the trader if $\pi(0) = -\infty$: by pure trading, the risk can be driven below every bound, and the hedging and pricing problem degenerates (every liability would have indifference price $-\infty$). Irrelevance is a joint property of the market *and* the risk measure, and it can occur without classical arbitrage.

Example 2.2. The following is a simplified version of the statistical-arbitrage illustration of Bühler et al. [2, Section 3.1]. We will show, that a market without arbitrage can still be irrelevant. Consider a single period, one asset, no costs and no constraints, and the risk measure $\rho(X) = -\mathbb{E}[X]$ of Section 2.4 below. Let the price increment be

$$S_1 - S_0 = 1/2 + \varepsilon,$$

where ε takes the values ± 1 with probability $1/2$ each. There is no arbitrage: every nonzero position loses money with probability $1/2$. But the strategy $\delta_0 = m$ has

$$\rho\big(W(\delta)\big) = -\mathbb{E}[m(1/2 + \varepsilon)] = -m/2 \longrightarrow -\infty \qquad (m \to \infty),$$

so $\pi(0) = -\infty$ leading to a irrelevent market. With a more risk averse p, the market could be relevant.

Relevance of the market is a condition on the position $X = 0$ only. It is natural to ask whether it already guarantees feasibility, meaning $\pi(X) > -\infty$, for all other positions. On a finite probability space, Bühler et al. [2, Corollary 3.3] answer this affirmatively: since π is monotone and cash-invariant (this is proved in Chapter 3), one has for every X

$$\pi(X) \geq \pi\big(\sup X\big) = \pi(0) - \sup X > -\infty,$$

because on a finite Ω every random variable satisfies $\sup X < \infty$. On a general probability space this argument does not survive for unbounded X. This is the first appearance of the finite-versus-general theme of this thesis, and an early hint of why bounded liabilities, hypothesis (A1) of Chapter 3, play a distinguished role.

2.4 Measuring risk: two naïve attempts

We now turn to the second half of the question: given the random P&L (2.1), how should one quantify its riskiness? We look for a functional ρ that maps a random position X to a number $\rho(X)$, interpreted as *the amount of cash that must be added to the position to make it acceptable.* A positive $\rho(X)$ is a capital requirement; a negative $\rho(X)$ means cash can be withdrawn without making the position unacceptable. Before giving the axioms, we examine two natural first candidates and discuss why they fail to be good risk measures. Their failures are instructive because each one points directly at one of the axioms below.

Example 2.3. Negative expectation. The map $\rho(X) = -\mathbb{E}[X]$ seems very natural in the cash interpretation: The exact amount of cash, that must be added to X so that the expected P&L becomes nonnegative, is also the amount that makes the position acceptable. Its defect is that it ignores the distribution of X entirely: a sure gain of 1 and a coin flip paying $+1000$ or -998 with probability $1/2$ each have the same expectation, hence the same risk number, though no risk-averse trader is indifferent between them. Example 2.2 showed that under the negative expectation the market gets irrelevant at the slightest presence of statistical arbitrage. Whatever ρ we choose, it must be able to distinguish positions by more than their mean.

Example 2.4. Variance. The map $\rho(X) = \mathrm{Var}(X)$ directly tackels the missing ingredient of the negative expectation: It punishes uncertainty. Furthermore, it is the canonical measure of risk in Markowitz portfolio theory [4]. Unfortunately, it still fails the cash interpretation twice. First, it penalizes gains and losses symmetrically: a position paying $+1000$ or $+998$ with probability $1/2$ has the same variance as one paying -1000 or -998, yet only the latter is risky in any economic sense. In particular the variance is not monotone: improving a position in every scenario can increase its risk number. Second, $\mathrm{Var}(X+c) = \mathrm{Var}(X)$ for every cash amount c, so adding cash does not reduce the risk number at all; the number cannot be a capital requirement, since no amount of capital ever makes the position acceptable. Both failures make economic nonsense of the reading of $\rho(X)$ as required capital.

These two failures motivate the axioms of the next section: monotonicity (better positions need less capital) and cash-invariance (added cash reduces the requirement one for one), together with convexity (diversification does not increase risk).

2.5 Convex risk measures

Let $\mathcal{X} = \{X : \Omega \to \mathbb{R}\}$ be a vector space of random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. Before defining the convex risk measure ρ, we need to discuss the choice of the domain space of random variables $\mathcal{X}$. On a finite probability space, the space question is redundant. Every random variable $X \in \mathcal{X}$ is finite-dimensional and ρ is a convex function on $\mathbb{R}^N$. Therefore, it is automatically bounded and continuous. On a general Ω one must choose $\mathcal{X}$, and the choice matters. E.g. taking $\mathcal{X} = L^\infty$ keeps ρ finite but is a very restrictive choice. The choice adopted in this thesis, following the needs of Chapter 3, is the largest one: ρ is defined on L^0, the space of all real-valued random variables. The following definition is that of Föllmer and Schied [3, Definition 4.1]; it is also the notion used by Bühler et al. [2, Definition 1].

Definition 2.5. A map $\rho\colon L^0 \to [-\infty, \infty]$ is a *convex risk measure* if for all $X, Y \in L^0$:

(i) *monotonicity:* $X \le Y$ $\mathbb{P}$-a.s. implies $\rho(X) \ge \rho(Y)$;

(ii) *cash-invariance:* $\rho(X + c) = \rho(X) - c$ for every $c \in \mathbb{R}$;

(iii) *convexity:* $\rho(\alpha X + (1-\alpha)Y) \le \alpha\rho(X) + (1-\alpha)\rho(Y)$ for every $\alpha \in [0, 1]$.

Each axiom has a direct economic reading, with the usual conventions of extended-real arithmetic when ρ takes the values $\pm\infty$. Monotonicity says that a position that pays more in every scenario needs no more capital. Cash-invariance says that capital already in the position counts toward the requirement one for one; it makes the unit of $\rho(X)$ the same as the unit of X, namely cash at time T. Whenever $\rho(X) \in \mathbb{R}$ is finite, cash-invariance gives

$$\rho\big(X + \rho(X)\big) = \rho(X) - \rho(X) = 0,$$

so $\rho(X)$ is exactly the minimal cash injection that makes the position acceptable in the sense of having nonpositive risk. If $\rho(X) = +\infty$, no finite cash amount renders the position acceptable; if $\rho(X) = -\infty$, the position is infinitely attractive and any finite cash amount can be withdrawn while keeping risk nonpositive. Convexity encodes the diversification principle: the risk of a mixture is at most the mixture of the risks, so splitting capital between two positions is never penalized. The negative expectation satisfies all three axioms (its failure in Section 2.4 is one of risk sensitivity, not of the axioms; it will be excluded later by the continuity hypothesis (A3) of Chapter 3), while the variance violates (i) and (ii).

2.6 Optimized certainty equivalents

The convex risk measures used in this thesis all arise from a single construction, the *optimized certainty equivalent* (OCE) of Ben-Tal and Teboulle [1], which Bühler et al. [2, Section 3.3] also take as their working class. The construction turns a one-dimensional loss function into a risk measure, and its variational form is what makes both the theory (Chapters 3 and 4) and the numerics (the training algorithm of Chapter 4) tractable.

Throughout this section, $\ell\colon \mathbb{R} \to \mathbb{R}$ is a *loss function*: convex, nondecreasing and finite on all of $\mathbb{R}$ (hence continuous). The value $\ell(x)$ is read as the disutility of a loss of size x; monotonicity says larger losses hurt more, convexity says they hurt disproportionately more.

Definition 2.6. The *optimized certainty equivalent* associated with ℓ is

$$\rho_\ell(X) := \inf_{w\in\mathbb{R}} \Big(w + \mathbb{E}\big[\ell(-X-w)\big] \Big), \tag{2.2}$$

for all X for which the expectation is well defined in $(-\infty, \infty]$ for every $w \in \mathbb{R}$.

The interpretation of (2.2) is a capital allocation: the trader sets aside the cash reserve w today and suffers the disutility $\ell(-X-w)$ of the shortfall of the reserved position; optimizing over w gives the best such allocation. The infimum over a single real parameter is what later turns the hedging problem into a finite-dimensional training problem.

For (2.2) to define a sensible risk measure, the loss function must be normalized. We adopt the normalization of Ben-Tal and Teboulle [1] as a standing assumption.

Assumption 2.7. The loss function ℓ satisfies $\ell(0) = 0$ and $1 \in \partial\ell(0)$, where $\partial\ell(0)$ denotes the subdifferential of ℓ at 0.

The normalization $\ell(0) = 0$ is a mere calibration of the zero level of disutility. The slope condition $1 \in \partial\ell(0)$ is essential, for the following reason. If every subgradient of ℓ were strictly below 1, say $\ell' \leq \gamma < 1$ everywhere, then for every fixed x,

$$w + \ell(-x-w) \leq w + \ell(-x) + \gamma|w| \longrightarrow -\infty \qquad (w \to -\infty),$$

so the infimum in (2.2) would equal $-\infty$ for *every* position X: reserving ever more negative amounts would be rewarded faster by the cash term than punished by the loss term, and the OCE would degenerate to $\rho_\ell \equiv -\infty$. Some slope of size at least 1 is therefore necessary; placing it at 0, together with $\ell(0) = 0$, is the convenient normalization. Stating Assumption 2.7 once and for all keeps every finiteness discussion in Chapters 3 and 4 clean; both examples at the end of this section satisfy it.

Remark 2.8. On a finite probability space, the expectation in (2.2) is a finite sum and the definition needs no discussion. On a general Ω, well-definedness must be checked — this is the second appearance of the finite-versus-general theme. Let $X \in L^1$. Decompose $\ell(-X - w)$ into positive and negative parts. The negative part is controlled by the affine minorant supplied by Assumption 2.7: convexity gives

$$\ell(x) \geq \ell(0) + 1 \cdot x = x \qquad \text{for all } x \in \mathbb{R},$$

so $\ell(-X - w) \geq -X - w$ and hence $\ell(-X - w)^- \leq |X| + |w| \in L^1$. The expectation $\mathbb{E}[\ell(-X - w)]$ is therefore well defined in $(-\infty, \infty]$ for every w, and (2.2) makes sense, with $\rho_\ell(X) = +\infty$ possible when the positive part fails to be integrable for every w. Moreover, the same minorant yields

$$w + \mathbb{E}\big[\ell(-X - w)\big] \geq w + \mathbb{E}[-X - w] = -\mathbb{E}[X],$$

so $\rho_\ell(X) \geq -\mathbb{E}[X] > -\infty$ for every $X \in L^1$: the OCE never takes the value $-\infty$ on L^1. On positions outside L^1 we use (2.2) with the conventions of extended-real arithmetic, consistently with the domain choice of Remark **??**.

The next lemma is the general-Ω version of Bühler et al. [2, Lemma 3.5]; the proof is essentially theirs, rewritten for extended-real expectations on L^1.

Lemma 2.9. *Under Assumption 2.7, the map ρ_ℓ defined by* (2.2) *is a convex risk measure on L^1, with values in $(-\infty, \infty]$.*

Proof. Finiteness from below was shown in Remark 2.8. We verify the three axioms of Definition 2.5.

Monotonicity. If $X_1 \leq X_2$ $\mathbb{P}$-a.s., then $-X_1 - w \geq -X_2 - w$ $\mathbb{P}$-a.s. for every w, so monotonicity of ℓ and of the expectation give

$$w + \mathbb{E}\big[\ell(-X_1 - w)\big] \geq w + \mathbb{E}\big[\ell(-X_2 - w)\big].$$

Taking the infimum over w on both sides yields $\rho_\ell(X_1) \geq \rho_\ell(X_2)$.

Cash-invariance. For $c \in \mathbb{R}$, the substitution $w' = w + c$ gives

$$\begin{aligned} \rho_\ell(X + c) &= \inf_{w \in \mathbb{R}} \Big(w + \mathbb{E}\big[\ell(-X - c - w)\big]\Big) \\ &= \inf_{w' \in \mathbb{R}} \Big(w' - c + \mathbb{E}\big[\ell(-X - w')\big]\Big) = \rho_\ell(X) - c. \end{aligned}$$

Convexity. Fix $\alpha \in [0,1]$, positions X_1, X_2 and reserves $w_1, w_2 \in \mathbb{R}$, and set $w_\alpha := \alpha w_1 + (1-\alpha) w_2$. Convexity of ℓ applied ω-wise gives

$$\ell\big(-\alpha X_1 - (1-\alpha)X_2 - w_\alpha\big) \le \alpha \ell(-X_1 - w_1) + (1-\alpha)\ell(-X_2 - w_2) \ \mathbb{P}\text{-a.s.}$$

Adding w_α, taking expectations and estimating the left-hand side from below by the infimum over w,

$$\begin{aligned} \rho_\ell\big(\alpha X_1 + (1-\alpha)X_2\big) \le \alpha\Big(&w_1 + \mathbb{E}\big[\ell(-X_1 - w_1)\big]\Big) \\ &+ (1-\alpha)\Big(w_2 + \mathbb{E}\big[\ell(-X_2 - w_2)\big]\Big). \end{aligned}$$

Taking the infimum over w_1 and w_2 separately yields convexity. All expectations used are well defined by Remark 2.8, and the extended-real conventions cause no harm since only the value $+\infty$ can occur. □

In the language of Ben-Tal and Teboulle [1], the quantity $-\rho_\ell(X)$ is the optimized certainty equivalent of the position X under the utility function $u(x) = -\ell(-x)$: the sure amount the trader considers equivalent to holding X, after optimally splitting off a cash reserve. The two normalizations correspond under this translation.

We close the chapter with the two families of OCEs that serve as running examples in the rest of the thesis. They sit at opposite ends of a risk-aversion dial and demonstrate the flexibility of the class.

Example 2.10. (Entropic risk measure.) For risk aversion $\lambda > 0$, take the loss function

$$\ell_\lambda(x) := \frac{e^{\lambda x} - 1}{\lambda},$$

which is convex, increasing and satisfies $\ell_\lambda(0) = 0$ and $\ell'_\lambda(0) = 1$, so Assumption 2.7 holds. Fix X with $\mathbb{E}[e^{-\lambda X}] < \infty$ and set

$$f(w) := w + \mathbb{E}\big[\ell_\lambda(-X - w)\big] = w + \frac{1}{\lambda}\Big(e^{-\lambda w}\mathbb{E}\big[e^{-\lambda X}\big] - 1\Big).$$

The function f is smooth and strictly convex with $f'(w) = 1 - e^{-\lambda w}\mathbb{E}[e^{-\lambda X}]$, so the unique minimizer is

$$w^* = \frac{1}{\lambda}\log \mathbb{E}\big[e^{-\lambda X}\big],$$

and since $e^{-\lambda w^*}\mathbb{E}[e^{-\lambda X}] = 1$, one gets $f(w^*) = w^*$. Hence

$$\rho_{\ell_\lambda}(X) = \frac{1}{\lambda}\log \mathbb{E}\big[e^{-\lambda X}\big],$$

the *entropic risk measure* with parameter λ; if $\mathbb{E}[e^{-\lambda X}] = \infty$, then $f \equiv \infty$ and $\rho_{\ell_\lambda}(X) = \infty$, consistently with the extended-real convention. Bühler et al. [2, Example 3.5] use the shifted loss function $x \mapsto e^{\lambda x} - (1 + \log \lambda)/\lambda$, which produces the same risk measure. The explicit minimizer w^* is a computational asset: in the training algorithm of Chapter 4, the reserve variable can be eliminated in closed form for the entropic case. The entropic risk measure is also the link to classical utility theory: by Bühler et al. [2, Lemma 3.4], hedging under ρ_{ℓ_λ} is equivalent to maximizing exponential utility with risk aversion λ, and the associated indifference prices coincide.

Example 2.11. (Conditional value at risk / expected shortfall.) For a level $\alpha \in [0, 1)$, take the loss function

$$\ell_\alpha(x) := \frac{\max(x, 0)}{1 - \alpha},$$

which is convex and non-decreasing with $\ell_\alpha(0) = 0$ and $\partial \ell_\alpha(0) = [0, 1/(1 - \alpha)] \ni 1$, so Assumption 2.7 holds. The resulting OCE is the *average value at risk* (also called conditional value at risk or expected shortfall) at level $1 - \alpha$,

$$\rho_{\ell_\alpha}(X) = \inf_{w \in \mathbb{R}} \Big(w + \frac{1}{1 - \alpha} \mathbb{E}\big[(-X - w)^+\big] \Big) = \mathrm{AV@R}_{1-\alpha}(X),$$

the average of the value at risk of X over all levels below $1 - \alpha$; see Föllmer and Schied [3, Section 4.4] and Ben-Tal and Teboulle [1, Example 2.3]. The level α acts as a risk-aversion dial. For $\alpha = 0$, the function $w \mapsto w + \mathbb{E}[(-X - w)^+]$ equals $-\mathbb{E}[X]$ for all $w \leq \operatorname{ess\,inf}(-X)$ and is non-decreasing beyond, so $\rho_{\ell_0}(X) = -\mathbb{E}[X]$: the risk-neutral criterion of Section 2.4. As $\alpha \to 1$, the penalty $1/(1 - \alpha)$ forces the reserve to cover the loss in almost every scenario, and $\rho_{\ell_\alpha}(X) \to -\operatorname{ess\,inf} X$: the worst-case criterion. In between, the trader interpolates between pricing by expectation and superhedging.

With the market of Sections 2.1–2.3 and the risk measures of Sections 2.5–2.6 in place, Chapter 3 assembles them into the hedging functional π and studies the existence of optimal hedging strategies.

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